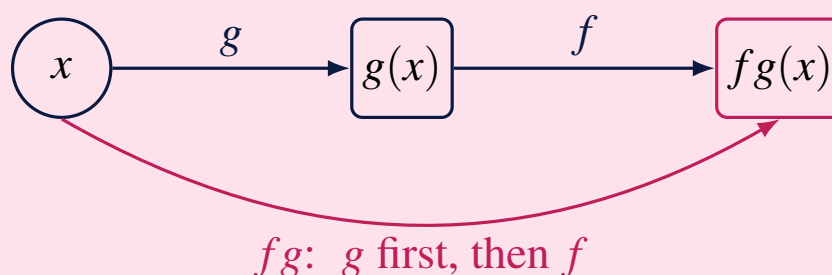


Composite & Inverse Functions

Notes · Questions · Worked Answers

Function notation · Composite functions $fg(x)$ · Inverse functions $f^{-1}(x)$ · Solving with functions



How to use this booklet

- Read the pink **Key Notes** pages first. The **Index** on the next page lists every question.
- Section 1–2: Pemberton exercises. Sections 3–4: Cambridge IGCSE 0580 exam questions, **most recent first**; the exam session, paper and question number are shown above each question.
- Each white **Question** page is followed by a yellow **Answer** page with working.

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Key Notes

1. Function notation

A **function** maps each input to exactly one output. $f(x) = 3x + 4$ can also be written $f : x \mapsto 3x + 4$.

- $f(2)$: replace x by 2: $f(2) = 3(2) + 4 = 10$.
- $f(x+2)$, $f(2x)$, $f(x^2)$: replace x by the whole bracket: $f(2x) = 3(2x) + 4 = 6x + 4$.
- “Find x when $f(x) = 25$ ”: solve $3x + 4 = 25 \Rightarrow x = 7$.

2. Composite functions

$fg(x)$ means $f(g(x))$: do g **first** (the one nearest x), then f .

$$\begin{array}{lll} f(x) = x^2, g(x) = x + 5 : & fg(3) = f(8) = 64 & gf(3) = g(9) = 14 \\ & fg(x) = f(x+5) = (x+5)^2 & gf(x) = x^2 + 5 \end{array}$$

- In general $fg(x) \neq gf(x)$. $ff(x)$ means $f(f(x))$.
- Three functions: $fgh(x) = f(g(h(x)))$ — work from the right.
- Simplify fully: expand brackets, collect like terms, or write as a single fraction.

3. Inverse functions

$f^{-1}(x)$ **undoes** f . If $f(3) = 10$ then $f^{-1}(10) = 3$.

Method: (1) write $y = f(x)$ (2) rearrange to make x the subject (3) swap x and y .

$$f(x) = \frac{3x+1}{4} : \quad y = \frac{3x+1}{4} \Rightarrow 4y = 3x+1 \Rightarrow x = \frac{4y-1}{3} \quad \Rightarrow \quad f^{-1}(x) = \frac{4x-1}{3}$$

4. Harder inverses

x appears twice (fractions):

$$y = \frac{2x+1}{x-3}$$

$$xy - 3y = 2x + 1$$

$$xy - 2x = 3y + 1$$

$$x(y-2) = 3y + 1$$

$$f^{-1}(x) = \frac{3x+1}{x-2}$$

Powers and roots:

$$f(x) = (x-1)^3 \Rightarrow f^{-1}(x) = \sqrt[3]{x} + 1$$

$$f(x) = x^3 + 1 \Rightarrow f^{-1}(x) = \sqrt[3]{x-1}$$

Exponentials: $h(x) = 2^x$: $h^{-1}(16) = 4$
because $2^4 = 16$.

5. Useful facts

- $ff^{-1}(x) = f^{-1}f(x) = x$, e.g. $ff^{-1}(7) = 7$ and $g^{-1}g(x+21) = x+21$.
- $f^{-1}(x) = a$ means $x = f(a)$ — you do **not** need to find f^{-1} first!
- A function is its own inverse when $f^{-1}(x) = f(x)$ (equivalently $ff(x) = x$), e.g. $\frac{1}{x}$, $1-x$.
- Solving $f(x) = f^{-1}(x)$: find f^{-1} , put them equal and solve.
- The graph of $y = f^{-1}(x)$ is the reflection of $y = f(x)$ in the line $y = x$.

6. Typical exam traps

- Order: $gf(2)$ means f first. Write the inner value down before applying the outer function.
- Use brackets when substituting: $f(x) = x^2 + 3x$, $f(2x-1) = (2x-1)^2 + 3(2x-1)$.
- $f^{-1}(x)$ is **not** $\frac{1}{f(x)}$.
- “Simplest form” / “single fraction”: common denominator, expand, collect terms.
- Indices: $64^x = 2^{6x}$, $\frac{1}{3^x} = 3^{-x}$, $\frac{1}{32} = 2^{-5}$.

1**Pemberton Ex 5.5 · Q1** $f(x) = 2x$ and $g(x) = x - 8$. Find **a** $fg(5)$ **b** $gf(5)$ **2****Pemberton Ex 5.5 · Q2** $f(x) = 3x + 1$ and $g(x) = 2 - 3x$. Find **a** $fg(4)$ **b** $gf(4)$ **3****Pemberton Ex 5.5 · Q3** $f(x) = x - 3$ and $g(x) = x^2$. Find **a** $fg(-2)$ **b** $gf(-2)$ **4****Pemberton Ex 5.5 · Q4** $f(x) = 5 + \frac{6}{x}$ and $g(x) = x - 4$. Find **a** $fg(1)$ **b** $gf(1)$

1 *Pemberton Ex 5.5 · Q1*

$$(a) g(5) = -3, f(-3) = \boxed{-6} \quad (b) f(5) = 10, g(10) = \boxed{2}$$

2 *Pemberton Ex 5.5 · Q2*

$$(a) g(4) = -10, f(-10) = \boxed{-29} \quad (b) f(4) = 13, g(13) = \boxed{-37}$$

3 *Pemberton Ex 5.5 · Q3*

$$(a) g(-2) = 4, f(4) = \boxed{1} \quad (b) f(-2) = -5, g(-5) = \boxed{25}$$

4 *Pemberton Ex 5.5 · Q4*

$$(a) g(1) = -3, f(-3) = 5 - 2 = \boxed{3} \quad (b) f(1) = 11, g(11) = \boxed{7}$$

5**Pemberton Ex 5.5 · Q5**

$f(x) = \frac{4}{x+1}$ and $g(x) = \frac{1}{x-3}$. Find **a** $fg(3)$ **b** $gf(3)$

6**Pemberton Ex 5.5 · Q6**

$f(x) = (x+1)^2$, $g(x) = 3-x$ and $h(x) = \frac{1}{x-4}$.

Find **a** $f(-4)$ **b** $gf(2)$ **c** $fh(6)$ **d** $fgh(5)$ **e** $ff(-2)$

7**Pemberton Ex 5.5 · Q7**

$p : x \rightarrow x-5$, $q : x \rightarrow \frac{x}{2}$ and $r : x \rightarrow x^2 + 3$.

Find **a** $q(-12)$ **b** $qr(5)$ **c** $pq(-4)$ **d** $pqr(6)$ **e** $qq(3)$

5 Pemberton Ex 5.5 · Q5

(a) $g(3) = \frac{1}{0}$ is **undefined**, so $fg(3)$ cannot be found. (b) $f(3) = 1$,
 $g(1) = \boxed{-\frac{1}{2}}$

6 Pemberton Ex 5.5 · Q6

(a) $\boxed{9}$ (b) $g(9) = \boxed{-6}$ (c) $h(6) = \frac{1}{2}$, $f(\frac{1}{2}) = \boxed{\frac{9}{4}}$ (d) $h(5) = 1$, $g(1) = 2$,
 $f(2) = \boxed{9}$ (e) $f(-2) = 1$, $f(1) = \boxed{4}$

7 Pemberton Ex 5.5 · Q7

(a) $\boxed{-6}$ (b) $r(5) = 28$, $q(28) = \boxed{14}$ (c) $q(-4) = -2$, $p(-2) = \boxed{-7}$ (d)
 $r(6) = 39$, $q = 19.5$, $p = \boxed{14.5}$ (e) $\frac{3}{2} \rightarrow \boxed{\frac{3}{4}}$

8**Pemberton Ex 5.5 · Q8**

If $f(x) = 2x + 3$ and $g(x) = 2x - 3$, match these cards.

a $fg(x)$ **b** $gf(x)$ **c** $gg(x)$ **d** $ff(x)$ **A** $4x + 3$ **B** $4x - 9$ **C** $4x + 9$ **D** $4x - 3$ **9****Pemberton Ex 5.5 · Q9**

$f(x) = x - 3$ and $g(x) = x + 8$. Find **a** $fg(x)$ **b** $gf(x)$ **c** $ff(x)$

10**Pemberton Ex 5.5 · Q10**

$p(x) = 2x + 1$ and $q(x) = x - 5$. Find **a** $qp(x)$ **b** $pq(x)$ **c** $pp(x)$

11**Pemberton Ex 5.5 · Q11**

$f(x) = x^2 + 3$ and $g(x) = 5x$. Find **a** $gf(x)$ **b** $fg(x)$ **c** $gg(x)$

8 *Pemberton Ex 5.5 · Q8*

$$fg = 2(2x - 3) + 3 = 4x - 3, gf = 2(2x + 3) - 3 = 4x + 3, gg = 2(2x - 3) - 3 = 4x - 9, ff = 2(2x + 3) + 3 = 4x + 9$$

$$a \rightarrow D, b \rightarrow A, c \rightarrow B, d \rightarrow C$$

9 *Pemberton Ex 5.5 · Q9*

$$(a) \boxed{x + 5} \quad (b) \boxed{x + 5} \quad (c) \boxed{x - 6}$$

10 *Pemberton Ex 5.5 · Q10*

$$(a) \boxed{2x - 4} \quad (b) 2(x - 5) + 1 = \boxed{2x - 9} \quad (c) 2(2x + 1) + 1 = \boxed{4x + 3}$$

11 *Pemberton Ex 5.5 · Q11*

$$(a) \boxed{5x^2 + 15} \quad (b) \boxed{25x^2 + 3} \quad (c) \boxed{25x}$$

12**Pemberton Ex 5.5 · Q12**

$p(x) = \frac{3}{x-2}$ and $q(x) = x + 1$. Find **a** $pq(x)$ **b** $qp(x)$

13**Pemberton Ex 5.5 · Q13**

$f(x) = x + 6$ and $g(x) = 3x$. Find x if $fg(x) = 27$.

14**Pemberton Ex 5.5 · Q14**

$f(x) = 2x - 3$ and $g(x) = \frac{x}{5}$. Find x if $gf(x) = 6$.

15**Pemberton Ex 5.5 · Q15**

$f(x) = x^2 + 2$ and $g(x) = \frac{1}{x-2}$. Find x if $gf(x) = 4$.

12 *Pemberton Ex 5.5 · Q12*

$$(a) \frac{3}{(x+1)-2} = \boxed{\frac{3}{x-1}} \quad (b) \frac{3}{x-2} + 1 = \boxed{\frac{x+1}{x-2}}$$

13 *Pemberton Ex 5.5 · Q13*

$$3x + 6 = 27 \Rightarrow \boxed{x = 7}$$

14 *Pemberton Ex 5.5 · Q14*

$$\frac{2x-3}{5} = 6 \Rightarrow 2x-3 = 30 \Rightarrow \boxed{x = 16.5}$$

15 *Pemberton Ex 5.5 · Q15*

$$gf(x) = \frac{1}{x^2} = 4 \Rightarrow x^2 = \frac{1}{4} \Rightarrow \boxed{x = \pm \frac{1}{2}}$$

16**Pemberton Ex 5.6 · Q1**

Write down the inverse of these functions.

a $f(x) = x + 2$ **b** $f(x) = x - 7$ **c** $f(x) = 22 + x$ **d** $f(x) = x - 0.5$
e $f(x) = 3x$ **f** $f(x) = 8x$ **g** $f(x) = \frac{x}{4}$ **h** $f(x) = \frac{x}{6}$

16 *Pemberton Ex 5.6 · Q1*

(a) $x - 2$ (b) $x + 7$ (c) $x - 22$ (d) $x + 0.5$ (e) $\frac{x}{3}$ (f) $\frac{x}{8}$ (g) $4x$ (h) $6x$

17

Pemberton Ex 5.6 · Q2

Find the inverse of these functions.

a $f(x) = 2x - 5$ **b** $f(x) = 3x + 8$ **c** $f(x) = 5x - 2$ **d** $f(x) = 3 + 4x$
e $f(x) = 3(x + 2)$ **f** $f(x) = 4(x - 7)$ **g** $f(x) = 2(x - 10)$ **h** $f(x) = 2(5 + x)$
i $f(x) = 2(5x + 4)$ **j** $f(x) = 3(5 - 2x)$ **k** $f(x) = 4(7 - 3x)$ **l** $f(x) = 3(2x - 9)$
m $f(x) = \frac{x - 2}{4}$ **n** $f(x) = \frac{x + 6}{5}$ **o** $f(x) = \frac{2x - 3}{4}$ **p** $f(x) = \frac{5 - 3x}{6}$

17 *Pemberton Ex 5.6 · Q2*

$$\begin{array}{llllll}
 \text{(a)} \frac{x+5}{2} & \text{(b)} \frac{x-8}{3} & \text{(c)} \frac{x+2}{5} & \text{(d)} \frac{x-3}{4} & \text{(e)} \frac{x-6}{3} & \text{(f)} \frac{x+28}{4} \\
 \text{(g)} \frac{x+20}{2} & \text{(h)} \frac{x-10}{2} & \text{(i)} \frac{x-8}{10} & \text{(j)} \frac{15-x}{6} & \text{(k)} \frac{28-x}{12} & \text{(l)} \frac{x+27}{6} \\
 \text{(m)} 4x+2 & \text{(n)} 5x-6 & \text{(o)} \frac{4x+3}{2} & \text{(p)} \frac{5-6x}{3}
 \end{array}$$

18**Pemberton Ex 5.6 • Q3**

Find the inverse of these functions.

$$\begin{array}{llll}
 \text{a } f(x) = \frac{5}{x} - 3 & \text{b } f(x) = \frac{3}{x} + 2 & \text{c } f(x) = \frac{6}{x} - 7 & \text{d } f(x) = 4 - \frac{5}{x} \\
 \text{e } f(x) = \frac{2}{3+x} & \text{f } f(x) = \frac{4}{2x+1} & \text{g } f(x) = \frac{3}{2x-5} & \text{h } f(x) = \frac{6}{3-2x} \\
 \text{i } f(x) = \frac{2x+1}{x-3} & \text{j } f(x) = \frac{x-7}{2x+5} & \text{k } f(x) = \frac{x+6}{x-8} & \text{l } f(x) = \frac{1-2x}{3-4x}
 \end{array}$$

19**Pemberton Ex 5.6 • Q4**

Find the inverse of these functions.

$$\text{a } f(x) = (x+3)^3 \quad \text{b } f(x) = (x-5)^3 \quad \text{c } f(x) = (2x+7)^3 \quad \text{d } f(x) = (5-3x)^3$$

18 Pemberton Ex 5.6 · Q3

$$\begin{array}{llllll}
 \text{(a)} \frac{5}{x+3} & \text{(b)} \frac{3}{x-2} & \text{(c)} \frac{6}{x+7} & \text{(d)} \frac{5}{4-x} & \text{(e)} \frac{2-3x}{x} & \text{(f)} \frac{4-x}{2x} \\
 \text{(g)} \frac{5x+3}{2x} & \text{(h)} \frac{3x-6}{2x} & \text{(i)} \frac{3x+1}{x-2} & \text{(j)} \frac{5x+7}{1-2x} & \text{(k)} \frac{8x+6}{x-1} & \text{(l)} \frac{1-3x}{2-4x}
 \end{array}$$

e.g. (i): $y = \frac{2x+1}{x-3} \Rightarrow xy - 3y = 2x + 1 \Rightarrow x(y-2) = 3y + 1 \Rightarrow x = \frac{3y+1}{y-2}$

19 Pemberton Ex 5.6 · Q4

$$\text{(a)} \sqrt[3]{x} - 3 \quad \text{(b)} \sqrt[3]{x} + 5 \quad \text{(c)} \frac{\sqrt[3]{x} - 7}{2} \quad \text{(d)} \frac{5 - \sqrt[3]{x}}{3}$$

20**Pemberton Ex 5.6 · Q5**

If $f(x) = 2x + 3$, find **a** $f^{-1}(0)$ **b** $f^{-1}(7)$ **c** $f^{-1}(2)$

21**Pemberton Ex 5.6 · Q6**

If $f(x) = 6 - 5x$, find **a** $f^{-1}(4)$ **b** $f^{-1}(-2)$ **c** $f^{-1}(6)$

22**Pemberton Ex 5.6 · Q7**

If $f(x) = 2x - 1$, solve the equation $f(x) = f^{-1}(x)$.

23**Pemberton Ex 5.6 · Q8**

If $f(x) = 3x + 2$, solve the equation $f(x) = f^{-1}(x)$.

20 Pemberton Ex 5.6 · Q5

$$f^{-1}(x) = \frac{x-3}{2}: \text{ (a) } \boxed{-1.5} \text{ (b) } \boxed{2} \text{ (c) } \boxed{-0.5}$$

21 Pemberton Ex 5.6 · Q6

$$f^{-1}(x) = \frac{6-x}{5}: \text{ (a) } \boxed{0.4} \text{ (b) } \boxed{1.6} \text{ (c) } \boxed{0}$$

22 Pemberton Ex 5.6 · Q7

$$2x - 1 = \frac{x+1}{2} \Rightarrow 4x - 2 = x + 1 \Rightarrow \boxed{x = 1}$$

23 Pemberton Ex 5.6 · Q8

$$3x + 2 = \frac{x-2}{3} \Rightarrow 9x + 6 = x - 2 \Rightarrow \boxed{x = -1}$$

24

Pemberton Ex 5.6 · Q9

Remember: $f^{-1}f(x) = ff^{-1}(x) = x$, because f^{-1} undoes what f has done.

a If $f(x) = 2x - 9$, write down the value of $f^{-1}f(3)$.

b If $f(x) = 2(5x - 11)$, write down the value of $f^{-1}f(2.5)$.

c If $f(x) = \frac{2x - 9}{3x + 4}$, write down the value of $ff^{-1}(-29)$.

24 *Pemberton Ex 5.6 · Q9*

(a) $\boxed{3}$ (b) $\boxed{2.5}$ (c) $\boxed{-29}$

25**Oct/Nov 2024 · Paper 21 · Q18**

$$g(x) = 4^{x+3}$$

- (a) Find x when $g(x) = 1$.

..... [1]

- (b) Find $g^{-1}\left(\frac{1}{16}\right)$.

..... [2]

26**May/June 2024 · Paper 23 · Q20**

$$f(x) = 3^x + 2$$

- (a) Find x when $f(x) = 245$.

$x =$ [2]

- (b) Find x when $f^{-1}(x) = 7$.

$x =$ [2]

27**May/June 2023 · Paper 22 · Q14**

$$f(x) = 5x + 2$$

Find $f^{-1}(x)$.

$f^{-1}(x) =$ [2]

25 Oct/Nov 2024 · Paper 21 · Q18

(a) $4^{x+3} = 1 = 4^0 \Rightarrow \boxed{x = -3}$ (b) $g^{-1}\left(\frac{1}{16}\right)$ is the x with $4^{x+3} = 4^{-2}$:
 $\boxed{-5}$

26 May/June 2024 · Paper 23 · Q20

(a) $3^x = 243 = 3^5 \Rightarrow \boxed{x = 5}$ (b) $f^{-1}(x) = 7 \Rightarrow x = f(7) = 3^7 + 2 =$
 $\boxed{2189}$

27 May/June 2023 · Paper 22 · Q14

$$y = 5x + 2 \Rightarrow x = \frac{y-2}{5}; \quad \boxed{f^{-1}(x) = \frac{x-2}{5}}$$

28

May/June 2023 · Paper 23 · Q20

$$f(x) = 6x - 7 \qquad g(x) = x^{-3}$$

- (a) Find $f(x+2)$.
Give your answer in its simplest form.

..... [2]

- (b) Find $f^{-1}(x)$.

$f^{-1}(x) =$ [2]

- (c) Find x when $g(x) = f(22)$.

$x =$ [2]

28 May/June 2023 · Paper 23 · Q20

(a) $6(x+2) - 7 = \boxed{6x+5}$ (b) $\boxed{f^{-1}(x) = \frac{x+7}{6}}$

(c) $f(22) = 125$; $x^{-3} = 125 \Rightarrow x^3 = \frac{1}{125} \Rightarrow \boxed{x = \frac{1}{5}}$

29

Oct/Nov 2022 · Paper 22 · Q17

$$f(x) = x^2$$

$$g(x) = \frac{x+5}{2}$$

$$h(x) = 7x - 3$$

(a) Find $f(-3)$.

..... [1]

(b) Find $g^{-1}(x)$.

$g^{-1}(x) =$ [2]

(c) Solve $gf(x) = hh^{-1}(63)$ where $x > 0$.

$x =$ [3]

29 Oct/Nov 2022 · Paper 22 · Q17

(a) $\boxed{9}$ (b) $\boxed{g^{-1}(x) = 2x - 5}$ (c) $hh^{-1}(63) = 63; \frac{x^2 + 5}{2} = 63 \Rightarrow x^2 = 121 \Rightarrow \boxed{x = 11} (x > 0)$

30

Oct/Nov 2022 · Paper 23 · Q19

$$f(x) = 5x - 3, \quad x > 1$$

$$g(x) = \frac{10}{x-2}, \quad x \neq 2$$

- (a) Find $gf(x)$.
Give your answer in its simplest form.

..... [2]

- (b) Find $g^{-1}(x)$.

$$g^{-1}(x) = \text{.....} [3]$$

- (c) Find $ff^{-1}(x-1)$.

..... [1]

30 Oct/Nov 2022 · Paper 23 · Q19

$$(a) \quad gf(x) = \frac{10}{5x-3-2} = \frac{10}{5x-5} = \boxed{\frac{2}{x-1}}$$

$$(b) \quad y = \frac{10}{x-2} \Rightarrow x-2 = \frac{10}{y}; \quad \boxed{g^{-1}(x) = \frac{10}{x} + 2 = \frac{2x+10}{x}} \quad (c)$$

$$ff^{-1}(x-1) = \boxed{x-1}$$

31

May/June 2022 · Paper 22 · Q19

$$f(x) = 7x - 8$$

$$g(x) = \frac{4}{x} + 5$$

$$h(x) = 2^x + 1$$

- (a) Find $f^{-1}(x)$.

$$f^{-1}(x) = \dots\dots\dots [2]$$

- (b) Find the value of x when $h(x) = g\left(\frac{1}{3}\right)$.

$$x = \dots\dots\dots [2]$$

31 *May/June 2022 · Paper 22 · Q19*

(a) $f^{-1}(x) = \frac{x+8}{7}$ (b) $g\left(\frac{1}{3}\right) = 12 + 5 = 17$; $2^x + 1 = 17 \Rightarrow 2^x = 16 \Rightarrow$
 $x = 4$

32

May/June 2022 · Paper 23 · Q19

$f(x) = kx^2$

$g(x) = \frac{1}{x}$

$h(x) = \frac{7x-2}{5}$

$j(x) = \frac{3-10x}{14}$

(a) $f(-5k) = 675$

Find the value of k .

$k = \dots\dots\dots [2]$

(b) Find $gh(x)$.

$\dots\dots\dots [1]$

(c) Find $h^{-1}(x) + j(x)$.

Give your answer in its simplest form.

$\dots\dots\dots [4]$

32 May/June 2022 · Paper 23 · Q19

$$\begin{aligned} \text{(a)} \quad k(-5k)^2 = 25k^3 = 675 &\Rightarrow k^3 = 27 \Rightarrow \boxed{k = 3} & \text{(b)} \quad \boxed{gh(x) = \frac{5}{7x-2}} \\ \text{(c)} \quad h^{-1}(x) = \frac{5x+2}{7}; \quad \frac{5x+2}{7} + \frac{3-10x}{14} &= \frac{10x+4+3-10x}{14} = \frac{7}{14} = \boxed{\frac{1}{2}} \end{aligned}$$

33

Oct/Nov 2021 · Paper 22 · Q20

$$f(x) = 2^{x-3}$$

$$g(x) = 2x - 1$$

$$h(x) = \frac{5}{x-4}$$

(a) Find $ff(6)$.

..... [2]

(b) Find $g^{-1}g(x+21)$.

..... [1]

(c) Find x when $f(x) = h(84)$.

$x =$ [2]

33 *Oct/Nov 2021 · Paper 22 · Q20*

(a) $f(6) = 2^3 = 8$, $ff(6) = f(8) = 2^5 = \boxed{32}$ (b) g^{-1} undoes g : $\boxed{x + 21}$

(c) $h(84) = \frac{5}{80} = \frac{1}{16} = 2^{-4}$; $2^{x-3} = 2^{-4} \Rightarrow x - 3 = -4 \Rightarrow \boxed{x = -1}$

34

May/June 2021 · Paper 22 · Q18

$$f(x) = x^2 - 25$$

$$g(x) = x + 4$$

Solve $fg(x+1) = gf(x)$.

$x = \dots\dots\dots$ [4]

34 *May/June 2021 · Paper 22 · Q18*

$$g(x+1) = x+5, \text{ so } fg(x+1) = (x+5)^2 - 25 = x^2 + 10x. \quad gf(x) = x^2 - 25 + 4 = x^2 - 21.$$

$$x^2 + 10x = x^2 - 21 \Rightarrow 10x = -21 \Rightarrow \boxed{x = -2.1}$$

35

Oct/Nov 2020 • Paper 21 • Q17

- (a) $f(x) = 3x^2 + a$ where a is an integer.
 $f(-2) = 19$

Find the value of a .

$$a = \dots\dots\dots [2]$$

- (b) $g(x) = 2x + 7$ $h(x) = 3x - 8$

- (i) Find $gh(x)$ in its simplest form.

$$\dots\dots\dots [2]$$

- (ii) Find $g^{-1}(x)$.

$$g^{-1}(x) = \dots\dots\dots [2]$$

35 *Oct/Nov 2020 · Paper 21 · Q17*

(a) $3(-2)^2 + a = 19 \Rightarrow 12 + a = 19 \Rightarrow \boxed{a = 7}$

(b)(i) $gh(x) = 2(3x - 8) + 7 = \boxed{6x - 9}$ (ii) $\boxed{g^{-1}(x) = \frac{x - 7}{2}}$

36

May/June 2020 · Paper 21 · Q14

(a) $f(x) = 4x + 3$ $g(x) = 5x - 4$

$$fg(x) = 20x + p$$

Find the value of p .

$$p = \dots\dots\dots [2]$$

(b) $h(x) = \frac{5x-1}{3}$

Find $h^{-1}(x)$.

$$h^{-1}(x) = \dots\dots\dots [3]$$

36 May/June 2020 · Paper 21 · Q14

(a) $fg(x) = 4(5x - 4) + 3 = 20x - 13 \Rightarrow \boxed{p = -13}$

(b) $y = \frac{5x - 1}{3} \Rightarrow 3y = 5x - 1 \Rightarrow x = \frac{3y + 1}{5}; \boxed{h^{-1}(x) = \frac{3x + 1}{5}}$

37

Oct/Nov 2015 · Paper 21 · Q21

$f(x) = x^3$

$g(x) = 3x - 5$

$h(x) = 2x + 1$

Work out

(a) $ff(2)$,*Answer(a)* [2](b) $gh(x)$ and simplify your answer,*Answer(b)* [2](c) $h^{-1}(x)$, the inverse of $h(x)$.*Answer(c)* $h^{-1}(x) =$ [2]

37 *Oct/Nov 2015 · Paper 21 · Q21*

(a) $f(8) = \boxed{512}$ (b) $3(2x + 1) - 5 = \boxed{6x - 2}$ (c) $\boxed{\frac{x - 1}{2}}$

38

2015 • Paper 2 • Q22

$$f(x) = 5x - 3$$

$$g(x) = x^2$$

- (a) Find $fg(-2)$.

Answer(a) [2]

- (b) Find $gf(x)$, in terms of x , in its simplest form.

Answer(b) [2]

- (c) Find $f^{-1}(x)$.

Answer(c) $f^{-1}(x) =$ [2]

38 2015 · Paper 2 · Q22

(a) $g(-2) = 4$, $f(4) = \boxed{17}$ (b) $(5x - 3)^2 = \boxed{25x^2 - 30x + 9}$ (c)

$\boxed{\frac{x+3}{5}}$

39

2015 • Paper 2 • Q23

$$f(x) = 5 - 3x$$

- (a) Find $f(6)$.

Answer(a) [1]

- (b) Find $f(x + 2)$.

Answer(b) [1]

- (c) Find $ff(x)$, in its simplest form.

Answer(c) [2]

- (d) Find $f^{-1}(x)$, the inverse of $f(x)$.

Answer(d) $f^{-1}(x) =$ [2]

39 2015 · Paper 2 · Q23

(a) $\boxed{-13}$ (b) $5 - 3(x + 2) = \boxed{-3x - 1}$ (c) $5 - 3(5 - 3x) = \boxed{9x - 10}$
(d) $\boxed{\frac{5 - x}{3}}$

40

2015 · Paper 2 · Q24

$$f(x) = 3x + 5 \qquad g(x) = x^2$$

- (a) Find $g(3x)$.

Answer(a) [1]

- (b) Find $f^{-1}(x)$, the inverse function.

Answer(b) $f^{-1}(x) =$ [2]

- (c) Find $ff(x)$.
Give your answer in its simplest form.

Answer(c) [2]

40 2015 · Paper 2 · Q24

(a) $\boxed{9x^2}$ (b) $\boxed{\frac{x-5}{3}}$ (c) $3(3x+5)+5 = \boxed{9x+20}$

41

2014 · Paper 2 · Q20

$$f(x) = 3x - 2 \qquad g(x) = \frac{2}{x+1}, \quad x \neq -1$$

(a) Find $gf(2)$.

Answer(a) [2]

(b) Solve $g(x) = 10$.

Answer(b) x = [2]

(c) Simplify.

$$f(2x) - f(x+2)$$

Answer(c) [3]

41 2014 · Paper 2 · Q20

(a) $f(2) = 4$, $g(4) = \boxed{0.4}$ (b) $\frac{2}{x+1} = 10 \Rightarrow \boxed{x = -0.8}$ (c) $(6x - 2) - (3x + 4) = \boxed{3x - 6}$

42

2014 · Paper 2 · Q16

$$f(x) = (x - 3)^2$$

$$g(x) = \frac{x - 1}{4}$$

$$h(x) = x^3$$

Find

(a) $hf(1)$,

Answer(a) [2]

(b) $g^{-1}(x)$,

Answer(b) $g^{-1}(x) =$ [2]

(c) $gh(x)$,

Answer(c) $gh(x) =$ [1]

(d) the solution to the equation $f(x) = 0$.

Answer(d) $x =$ [1]

42 2014 · Paper 2 · Q16

(a) $f(1) = 4$, $h(4) = \boxed{64}$ (b) $\boxed{4x + 1}$ (c) $\boxed{\frac{x^3 - 1}{4}}$ (d) $\boxed{x = 3}$

43

2013 • Paper 2 • Q21

$$f(x) = 5x + 4 \qquad g(x) = \frac{1}{2x}, \quad x \neq 0 \qquad h(x) = \left(\frac{1}{2}\right)^x$$

Find

(a) $fg(5)$,*Answer(a)* [2](b) $gg(x)$ in its simplest form,*Answer(b)* $gg(x) =$ [2](c) $f^{-1}(x)$,*Answer(c)* $f^{-1}(x) =$ [2](d) the value of x when $h(x) = 8$.*Answer(d)* $x =$ [2]

43 2013 · Paper 2 · Q21

(a) $g(5) = 0.1, f(0.1) = \boxed{4.5}$ (b) $g\left(\frac{1}{2x}\right) = \frac{1}{2 \cdot \frac{1}{2x}} = \boxed{x}$ (c) $\boxed{\frac{x-4}{5}}$
(d) $\left(\frac{1}{2}\right)^x = 8 = \left(\frac{1}{2}\right)^{-3} \Rightarrow \boxed{x = -3}$

44

2013 • Paper 2 • Q19

$$f(x) = 2x + 3 \quad g(x) = x^2$$

(a) Find $fg(6)$.

Answer(a) [2]

(b) Solve the equation $gf(x) = 100$.

Answer(b) $x =$ or $x =$ [3]

(c) Find $f^{-1}(x)$.

Answer(c) $f^{-1}(x) =$ [2]

(d) Find $ff^{-1}(5)$.

Answer(d) [1]

44 2013 · Paper 2 · Q19

(a) $g(6) = 36$, $f(36) = \boxed{75}$ (b) $(2x + 3)^2 = 100 \Rightarrow 2x + 3 = \pm 10$:

$\boxed{x = 3.5 \text{ or } -6.5}$

(c) $\boxed{\frac{x-3}{2}}$ (d) $\boxed{5}$

45

Oct/Nov 2012 · Paper 21 · Q20

$$f(x) = 4(x + 1) \qquad g(x) = \frac{x^3}{2} - 1$$

- (a) Write down the value of x when $f^{-1}(x) = 2$.

Answer(a) $x =$ [1]

- (b) Find $fg(x)$. Give your answer in its simplest form.

Answer(b) $fg(x) =$ [2]

- (c) Find $g^{-1}(x)$.

Answer(c) $g^{-1}(x) =$ [3]

45 *Oct/Nov 2012 · Paper 21 · Q20*

(a) $x = f(2) = \boxed{12}$ (b) $f(g(x)) = 4\left(\frac{x^3}{2} - 1 + 1\right) = \boxed{2x^3}$ (c) $\boxed{\sqrt[3]{2(x+1)}}$

46

Oct/Nov 2012 · Paper 23 · Q23

$$f(x) = 3x + 5 \quad g(x) = 4x - 1$$

- (a) Find the value of $gg(3)$.

Answer(a) [2]

- (b) Find $fg(x)$, giving your answer in its simplest form.

Answer(b) $fg(x) =$ [2]

- (c) Solve the equation.

$$f^{-1}(x) = 11$$

Answer(c) $x =$ [1]

46 *Oct/Nov 2012 · Paper 23 · Q23*

(a) $g(3) = 11$, $g(11) = \boxed{43}$ (b) $3(4x - 1) + 5 = \boxed{12x + 2}$ (c) $x =$
 $f(11) = \boxed{38}$

47

May/June 2012 · Paper 22 · Q18

$$f(x) = (x + 2)^3 - 5$$

$$g(x) = 2x + 10$$

$$h(x) = \frac{1}{x}, \quad x \neq 0$$

Find

(a) $gf(x)$,

Answer(a) $gf(x) = \dots\dots\dots$ [2]

(b) $f^{-1}(x)$,

Answer(b) $f^{-1}(x) = \dots\dots\dots$ [3]

(c) $gh(-\frac{1}{5})$.

Answer(c) $\dots\dots\dots$ [2]

47 *May/June 2012 · Paper 22 · Q18*

(a) $\boxed{2(x+2)^3}$ ($= 2x^3 + 12x^2 + 24x + 16$) (b) $\boxed{\sqrt[3]{x+5}-2}$ (c)

$h(-\frac{1}{5}) = -5, g(-5) = \boxed{0}$

48

Oct/Nov 2011 · Paper 21 · Q17

$$f(x) = \frac{1}{x+4} \quad (x \neq -4)$$

$$g(x) = x^2 - 3x$$

$$h(x) = x^3 + 1$$

(a) Work out $fg(1)$.

Answer(a) [2]

(b) Find $h^{-1}(x)$.

Answer(b) $h^{-1}(x) =$ [2]

(c) Solve the equation $g(x) = -2$.

Answer(c) $x =$ or $x =$ [3]

48 *Oct/Nov 2011 · Paper 21 · Q17*

(a) $g(1) = -2$, $f(-2) = \boxed{\frac{1}{2}}$ (b) $\boxed{\sqrt[3]{x-1}}$ (c) $x^2 - 3x + 2 = 0 \Rightarrow$
 $\boxed{x = 1 \text{ or } 2}$

49

May/June 2011 · Paper 21 · Q20

$$f(x) = x^3$$

$$g(x) = 2x - 3$$

(a) Find

(i) $g(6)$,

Answer(a)(i) [1]

(ii) $f(2x)$.

Answer(a)(ii) [1]

(b) Solve $fg(x) = 125$.

Answer(b) $x =$ [3]

(c) Find the inverse function $g^{-1}(x)$.

Answer(c) $g^{-1}(x) =$ [2]

49 *May/June 2011 · Paper 21 · Q20*

(a)(i) $\boxed{9}$ (ii) $\boxed{8x^3}$ (b) $(2x - 3)^3 = 125 \Rightarrow 2x - 3 = 5 \Rightarrow \boxed{x = 4}$ (c)

$$\boxed{\frac{x+3}{2}}$$

50

May/June 2011 · Paper 22 · Q19

$$f(x) = x^2 \quad g(x) = 2^x \quad h(x) = 2x - 3$$

(a) Find $g(3)$.

Answer(a) [1]

(b) Find $hh(x)$ in its simplest form.

Answer(b) [2]

(c) Find $fg(x + 1)$ in its simplest form.

Answer(c) [2]

50 *May/June 2011 · Paper 22 · Q19*

(a) $\boxed{8}$ (b) $2(2x - 3) - 3 = \boxed{4x - 9}$ (c) $(2^{x+1})^2 = \boxed{2^{2x+2}}$ ($= 4^{x+1}$)

51

2011 • Paper 2 • Q25

$$f: x \rightarrow 2x - 7 \qquad g: x \rightarrow \frac{1}{x}$$

Find

(a) $fg\left(\frac{1}{2}\right),$

Answer(a) [2]

(b) $gf(x),$

Answer(b) $gf(x) =$ [1]

(c) $f^{-1}(x).$

Answer(c) $f^{-1}(x) =$ [2]

51 2011 · Paper 2 · Q25

(a) $g(\frac{1}{2}) = 2$, $f(2) = \boxed{-3}$ (b) $\boxed{\frac{1}{2x-7}}$ (c) $\boxed{\frac{x+7}{2}}$

52

May/June 2010 · Paper 21 · Q20

$$f(x) = (x - 1)^3 \qquad g(x) = (x - 1)^2 \qquad h(x) = 3x + 1$$

(a) Work out $fg(-1)$.

Answer(a) [2]

(b) Find $gh(x)$ in its simplest form.

Answer(b) [2]

(c) Find $f^{-1}(x)$.

Answer(c) [2]

52 *May/June 2010 · Paper 21 · Q20*

(a) $g(-1) = 4$, $f(4) = \boxed{27}$ (b) $(3x + 1 - 1)^2 = \boxed{9x^2}$ (c) $\boxed{\sqrt[3]{x} + 1}$

53

May/June 2010 · Paper 22 · Q18

(a) $f(x) = 1 - 2x$.

(i) Find $f(-5)$.

Answer(a)(i) [1]

(ii) $g(x) = 3x - 2$.

Find $gf(x)$. Simplify your answer.

Answer(a)(ii) [2]

(b) $h(x) = x^2 - 5x - 11$.

Solve $h(x) = 0$.

Show all your working and give your answer correct to 2 decimal places.

Answer(b) $x =$ or $x =$ [4]

53 May/June 2010 · Paper 22 · Q18

(a)(i) $\boxed{11}$ (ii) $3(1 - 2x) - 2 = \boxed{1 - 6x}$ (b) $x = \frac{5 \pm \sqrt{69}}{2}$: $\boxed{6.65 \text{ or } -1.65}$

54**Oct/Nov 2024 • Paper 41 • Q8(a)**

(a) $f(x) = 7 - 3x$ $g(x) = x^2 - 16$

(i) Find the values of x when $g(x) = 20$.

$$x = \dots\dots\dots \text{ or } x = \dots\dots\dots [2]$$

(ii) Find $f^{-1}(x)$.

$$f^{-1}(x) = \dots\dots\dots [2]$$

(iii) Find $gf(x) + 1$, giving your answer in its simplest form.

$$\dots\dots\dots [3]$$

55**Oct/Nov 2024 • Paper 42 • Q9**

$f(x) = 2x - 5$ $g(x) = x^2 - 2x$

(a) Find

(i) $f(7)$

$$\dots\dots\dots [1]$$

(ii) $gf(7)$

$$\dots\dots\dots [1]$$

(iii) $f^{-1}(x)$.

$$f^{-1}(x) = \dots\dots\dots [2]$$

(b) Find $gf(x) - 3g(x)$.

Give your answer in the form $ax^2 + bx + c$.

$$\dots\dots\dots [4]$$

54 Oct/Nov 2024 · Paper 41 · Q8(a)

(i) $x^2 = 36 \Rightarrow$ $x = 6$ or -6 (ii) $\frac{7-x}{3}$ (iii) $(7-3x)^2 - 16 + 1 =$
 $9x^2 - 42x + 34$

55 Oct/Nov 2024 · Paper 42 · Q9

(a)(i) 9 (ii) 63 (iii) $\frac{x+5}{2}$ (b) $(2x-5)^2 - 2(2x-5) - 3(x^2-2x) =$
 $x^2 - 18x + 35$

56

Oct/Nov 2024 · Paper 43 · Q11

$f(x) = 2x + 5$

$g(x) = 1 - 2x$

$h(x) = \frac{1}{x+1}, x \neq -1$

$j(x) = 2^x$

- (a) Find
- $g(-3)$
- .

..... [1]

- (b) Find
- $f(x)g(x) + fg(x) + 1$
- .
-
- Give your answer in its simplest form.

..... [4]

- (c) Find
- $g^{-1}(x)$
- .

$g^{-1}(x) = \dots\dots\dots [2]$

- (d) Find
- $hh(1)$
- .

..... [2]

- (e) Simplify
- $\frac{1}{f(x)} - h(x)$
- .

Give your answer as a single fraction in its simplest form.

..... [3]

- (f) Find
- x
- when
- $j(x) = \frac{1}{32}$
- .

$x = \dots\dots\dots [1]$

- (g) Find
- x
- when
- $j^{-1}(x) = 0$
- .

$x = \dots\dots\dots [1]$

56 Oct/Nov 2024 · Paper 43 · Q11

(a) $\boxed{7}$ (b) $(2x + 5)(1 - 2x) + 2(1 - 2x) + 5 + 1 = \boxed{-4x^2 - 12x + 13}$

(c) $\boxed{\frac{1-x}{2}}$

(d) $h(1) = \frac{1}{2}, h(\frac{1}{2}) = \boxed{\frac{2}{3}}$ (e) $\boxed{\frac{-x-4}{(2x+5)(x+1)}}$ (f) $\boxed{-5}$ (g) $\boxed{1}$

57

May/June 2024 · Paper 41 · Q9

$$f(x) = 4x + 1 \qquad g(x) = 6 - 2x \qquad h(x) = 3^{x-2}$$

(a) Find

(i) $f(3)$

..... [1]

(ii) $gf(3)$.

..... [1]

(b) Find $g^{-1}(x)$.

$g^{-1}(x) = \dots\dots\dots [2]$

(c) Find x when $f(x) = g(2x - 7)$.

$x = \dots\dots\dots [4]$

(d) Find the value of $hh(2)$.

..... [2]

(e) Find x when $h^{-1}(x) = 10$.

$x = \dots\dots\dots [2]$

57 May/June 2024 · Paper 41 · Q9

(a)(i) $\boxed{13}$ (ii) $\boxed{-20}$ (b) $\boxed{\frac{6-x}{2}}$ (c) $4x + 1 = 20 - 4x \Rightarrow \boxed{x = \frac{19}{8}}$
(d) $h(2) = 1, h(1) = \boxed{\frac{1}{3}}$ (e) $x = 3^8 = \boxed{6561}$

58

May/June 2024 · Paper 42 · Q7(d)

(d) $f(x) = \frac{1}{4x-1}, x \neq \frac{1}{4}$ $g(x) = 3^x$

(i) Find $f(4)$.

..... [1]

(ii) Find $gg(2)$.

..... [2]

(iii) Find k when $g(k) = f(7)$.

..... [2]

58 *May/June 2024 · Paper 42 · Q7(d)*

(i) $\boxed{\frac{1}{15}}$ (ii) $g(9) = \boxed{19683}$ (iii) $f(7) = \frac{1}{27} = 3^{-3} \Rightarrow \boxed{k = -3}$

59**Oct/Nov 2023 · Paper 41 · Q9**

$$f(x) = (3x + 1)(x + 5)(x - 4) \quad g(x) = 2x - 3 \quad h(x) = 4^{2x-1}$$

(a) Find

(i) $f(0)$

..... [1]

(ii) $g^{-1}(x)$

$g^{-1}(x) =$ [2]

(iii) $gh(2)$.

..... [2]

(b) $g(2x) = 7$

Find the value of x .

$x =$ [2]

(c) Simplify $g(x^2) + gg(x) + 1$.

..... [3]

(d) Find $h^{-1}(16)$.

Question continues on the next page →

..... [2]

(e) $f(x) = (3x + 1)(x + 5)(x - 4)$

This can be written in the form $f(x) = ax^3 + bx^2 + cx + d$.

Find the value of each of a , b , c and d .

$a = \dots\dots\dots b = \dots\dots\dots c = \dots\dots\dots d = \dots\dots\dots$ [3]

59 Oct/Nov 2023 · Paper 41 · Q9

- (a)(i) $\boxed{-20}$ (ii) $\boxed{\frac{x+3}{2}}$ (iii) $h(2) = 64, g(64) = \boxed{125}$ (b) $4x - 3 = 7 \Rightarrow$
 $\boxed{x = 2.5}$
- (c) $2x^2 - 3 + 2(2x - 3) - 3 + 1 = \boxed{2x^2 + 4x - 11}$ (d) $4^{2x-1} = 16 \Rightarrow \boxed{1.5}$
- (e) $\boxed{a = 3, b = 4, c = -59, d = -20}$

60

Oct/Nov 2023 · Paper 42 · Q11

$f(x) = 1 - 3x$

$g(x) = (x-1)^2$

$h(x) = \frac{3}{x}, x \neq 0$

- (a) Find
- $g(3)$
- .

..... [1]

- (b) Find
- $f(x-2)$
- , giving your answer in its simplest form.

..... [2]

- (c) Find
- $f^{-1}(x)$
- .

$f^{-1}(x) = \dots\dots\dots [2]$

- (d)
- $gf(x) - g(x)f(x) = 3x^3 + ax^2 + bx + c$

Find the value of each of a , b and c .

$a = \dots\dots\dots$

$b = \dots\dots\dots$

$c = \dots\dots\dots [5]$

- (e) Find
- $h(x) - f(x)$
- , giving your answer as a single fraction in its simplest form.

Question continues on the next page →

..... [3]

(f) $h(x^n) = 3x^7$

Find the value of n .

 $n =$ [1]

60 *Oct/Nov 2023 · Paper 42 · Q11*

(a) $\boxed{4}$ (b) $\boxed{7 - 3x}$ (c) $\boxed{\frac{1-x}{3}}$ (d) $9x^2 - (x-1)^2(1-3x):$

$$\boxed{a = 2, b = 5, c = -1}$$

(e) $\boxed{\frac{3x^2 - x + 3}{x}}$ (f) $\frac{3}{x^n} = 3x^7 \Rightarrow \boxed{n = -7}$

61

Oct/Nov 2023 · Paper 43 · Q6

$$f(x) = 5x - 3 \qquad g(x) = 64^x \qquad h(x) = \frac{2}{x+1}, \quad x \neq -1$$

(a) Find the value of

(i) $f(2)$

..... [1]

(ii) $gf(0.5)$.

..... [2]

(b) Find $h^{-1}(x)$.

$h^{-1}(x) = \dots\dots\dots [3]$

(c) Find x when $g(x) = \frac{1}{2^5}$.

$x = \dots\dots\dots [2]$

(d) Write as a single fraction in its simplest form $\frac{1}{f(x)} - h(x)$.

..... [4]

61 *Oct/Nov 2023 · Paper 43 · Q6*

(a)(i) $\boxed{7}$ (ii) $f(0.5) = -0.5$, $64^{-1/2} = \boxed{\frac{1}{8}}$ (b) $\boxed{h^{-1}(x) = \frac{2-x}{x}}$

(c) $2^{6x} = 2^{-5} \Rightarrow \boxed{x = -\frac{5}{6}}$ (d) $\frac{x+1-2(5x-3)}{(5x-3)(x+1)} = \boxed{\frac{7-9x}{(5x-3)(x+1)}}$

62

May/June 2023 · Paper 41 · Q10

$f(x) = x - 4$

$g(x) = 2x + 5$

$h(x) = 3^x$

(a) Find

(i) $f(-3)$

..... [1]

(ii) $g^{-1}(x)$

$g^{-1}(x) = \dots\dots\dots [2]$

(iii) $f(x) \times g(x) \times f(x)$.

..... [4]

(b) Find x when $h(x) = g(f(2))$.

$x = \dots\dots\dots [2]$

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(a)(i) $\boxed{-7}$ (ii) $\boxed{\frac{x-5}{2}}$ (iii) $(x-4)^2(2x+5) = \boxed{2x^3 - 11x^2 - 8x + 80}$

(b) $g(f(2)) = g(-2) = 1$; $3^x = 1 \Rightarrow \boxed{x = 0}$

63

Feb/March 2023 · Paper 42 · Q11

$$f(x) = 2x - 1 \quad g(x) = 3x + 2 \quad h(x) = \frac{1}{x}, x \neq 0 \quad j(x) = x^2$$

- (a) Find
- $j(-1)$
- .

..... [1]

- (b) Find
- x
- when
- $f(x) + g(x) = 0$
- .

 $x =$ [2]

- (c) Find
- $gg(x)$
- , giving your answer in its simplest form.

..... [2]

- (d) Find
- $hf(x) + gh(x)$
- , giving your answer as a single fraction in its simplest form.

..... [4]

- (e) When
- $pp(x) = x$
- ,
- $p(x)$
- is a function such that
- $p^{-1}(x) = p(x)$
- .

Draw a ring around the function that has this property.

$$f(x) = 2x - 1 \quad g(x) = 3x + 2 \quad h(x) = \frac{1}{x}, x \neq 0 \quad j(x) = x^2$$

[1]

63 *Feb/March 2023 · Paper 42 · Q11*

(a) $\boxed{1}$ (b) $5x + 1 = 0 \Rightarrow \boxed{x = -\frac{1}{5}}$ (c) $3(3x + 2) + 2 = \boxed{9x + 8}$

(d) $\frac{1}{2x-1} + \frac{3}{x} + 2 = \boxed{\frac{4x^2 + 5x - 3}{x(2x-1)}}$ (e) $\boxed{h(x)}$

64

Oct/Nov 2022 • Paper 41 • Q7

$f(x) = 10 - x$

$g(x) = \frac{2}{x}, x \neq 0$

$h(x) = 2^x$

$j(x) = 5 - 2x$

(a) (i) Find $g\left(\frac{1}{2}\right)$.

..... [1]

(ii) Find $hg\left(\frac{1}{2}\right)$.

..... [1]

(b) Find x when $f(x) = 7$. $x =$ [1](c) Find x when $g(x) = h(3)$. $x =$ [2](d) Find $j^{-1}(x)$. $j^{-1}(x) =$ [2](e) Write $f(x) + g(x) + 1$ as a single fraction in its simplest form.

..... [3]

(f) $(f(x))^2 - ff(x) = ax^2 + bx + c$ Find the values of a , b and c .

Question continues on the next page →

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

$$c = \dots\dots\dots [4]$$

(g) Find x when $h^{-1}(x) = 10$.

$$x = \dots\dots\dots [2]$$

64 *Oct/Nov 2022 · Paper 41 · Q7*

(a)(i) $\boxed{4}$ (ii) $2^4 = \boxed{16}$ (b) $\boxed{3}$ (c) $\frac{2}{x} = 8 \Rightarrow \boxed{x = \frac{1}{4}}$ (d) $\boxed{\frac{5-x}{2}}$

(e) $\boxed{\frac{-x^2 + 11x + 2}{x}}$ (f) $(10-x)^2 - x = x^2 - 21x + 100$: $\boxed{a = 1, b = -21, c = 100}$

(g) $\boxed{1024}$

65

May/June 2022 · Paper 41 · Q4

$f(x) = 2x - 1$

$g(x) = 3x - 2$

$h(x) = \frac{1}{x}, x \neq 0$

$j(x) = 5^x$

(a) Find

(i) $f(2)$,

..... [1]

(ii) $gf(2)$.

..... [1]

(b) Find $g^{-1}(x)$.

$g^{-1}(x) = \dots\dots\dots [2]$

(c) Find x when $h(x) = j(-2)$.

$x = \dots\dots\dots [2]$

(d) Write $f(x) - h(x)$ as a single fraction.

..... [2]

(e) Find the value of $jj(2)$.

..... [1]

(f) Find x when $j^{-1}(x) = 4$.

$x = \dots\dots\dots [2]$

65 *May/June 2022 · Paper 41 · Q4*

(a)(i) $\boxed{3}$ (ii) $g(3) = \boxed{7}$ (b) $\boxed{\frac{x+2}{3}}$ (c) $j(-2) = \frac{1}{25} \Rightarrow \boxed{x = 25}$

(d) $\boxed{\frac{2x^2 - x - 1}{x}}$ (e) $5^{25} = \boxed{2.98 \times 10^{17}}$ (f) $x = 5^4 = \boxed{625}$

66**Oct/Nov 2021 · Paper 41 · Q8(a)**

(a) $f(x) = 3 - 5x$

(i) Find x when $f(x) = -5$.

$$x = \dots\dots\dots [2]$$

(ii) Find $f^{-1}(x)$.

$$f^{-1}(x) = \dots\dots\dots [2]$$

66 *Oct/Nov 2021 · Paper 41 · Q8(a)*

(i) $3 - 5x = -5 \Rightarrow \boxed{x = 1.6}$ (ii) $\boxed{f^{-1}(x) = \frac{3-x}{5}}$

67

Oct/Nov 2021 · Paper 43 · Q11

$f(x) = 2x - 1$

$g(x) = x^2 + 2x$

$h(x) = 4^x$

$j(x) = 2^x$

(a) Find the value of

(i) $h(3)$,

..... [1]

(ii) $fh(3)$.

..... [1]

(b) Solve the equation $gf(x) = 0$. $x = \dots\dots\dots$ or $x = \dots\dots\dots$ [4](c) $p^{-1}(x) = f(x)$ Find $p(x)$.

..... [2]

(d) $h(x)j(x) = \frac{1}{\sqrt{2}}$ Find the value of x . $x = \dots\dots\dots$ [3]

67 *Oct/Nov 2021 · Paper 43 · Q11*

(a)(i) $\boxed{64}$ (ii) $f(64) = \boxed{127}$

(b) $(2x - 1)^2 + 2(2x - 1) = (2x - 1)(2x + 1) = 0 \Rightarrow \boxed{x = \frac{1}{2} \text{ or } -\frac{1}{2}}$

(c) $p = f^{-1}$: $\boxed{p(x) = \frac{x+1}{2}}$ (d) $2^{2x} \cdot 2^x = 2^{3x} = 2^{-1/2} \Rightarrow \boxed{x = -\frac{1}{6}}$

68

May/June 2021 • Paper 41 • Q12

$f(x) = 3 - 2x$

$g(x) = x^2 + 5$

$h(x) = x^3$

- (a) Find
- $f(-5)$
- .

..... [1]

- (b) Find
- $ff(x)$
- .

Give your answer in its simplest form.

..... [2]

- (c) Solve
- $g(x) = f(x) + 37$
- .

 $x = \dots\dots\dots$ or $x = \dots\dots\dots$ [4]

- (d) Find
- $f^{-1}(x)$
- .

 $f^{-1}(x) = \dots\dots\dots$ [2]

- (e) Find
- $hf(x) + g(x)$
- .

Give your answer in its simplest form.

..... [5]

68 *May/June 2021 · Paper 41 · Q12*

(a) $\boxed{13}$ (b) $3 - 2(3 - 2x) = \boxed{4x - 3}$

(c) $x^2 + 5 = 40 - 2x \Rightarrow x^2 + 2x - 35 = 0 \Rightarrow (x + 7)(x - 5) = 0:$

$\boxed{x = 5 \text{ or } -7}$ (d) $\boxed{\frac{3 - x}{2}}$

(e) $(3 - 2x)^3 + x^2 + 5 = \boxed{-8x^3 + 37x^2 - 54x + 32}$

69

May/June 2021 · Paper 43 · Q10

$f(x) = 3x - 2$

$g(x) = 5x - 7$

$h(x) = x^2 + x$

$j(x) = 3^x$

(a) Find

(i) $f(2)$,

..... [1]

(ii) $g(2)$,

..... [1]

(iii) $gf(2)$.

..... [1]

(b) Find $f^{-1}(x)$.

$f^{-1}(x) = \dots\dots\dots [2]$

(c) Find $hf(x)$, giving your answer in the form $ax^2 + bx + c$.

..... [3]

(d) Find the derivative of $h(x)$.

..... [1]

(e) (i) Find x when $j^{-1}(x) = 4$.

$x = \dots\dots\dots [1]$

(ii) Simplify $j^{-1}j(x)$.

..... [1]

69 May/June 2021 · Paper 43 · Q10

(a)(i) $\boxed{4}$ (ii) $\boxed{3}$ (iii) $g(4) = \boxed{13}$ (b) $\boxed{\frac{x+2}{3}}$

(c) $(3x-2)^2 + (3x-2) = \boxed{9x^2 - 9x + 2}$ (d) $\boxed{2x+1}$ (e)(i) $x = 3^4 =$

$\boxed{81}$ (ii) $\boxed{x}$

70

Oct/Nov 2020 · Paper 42 · Q10

$$f(x) = x^2 + 1 \quad g(x) = 1 - 2x \quad h(x) = \frac{1}{x}, \quad x \neq 0 \quad j(x) = 5^x$$

(a) Find the value of

(i) $f(3)$,

..... [1]

(ii) $gf(3)$.

..... [1]

(b) Find $g^{-1}(x)$.

$$g^{-1}(x) = \dots\dots\dots [2]$$

(c) Find x when $h(x) = 2$.

$$x = \dots\dots\dots [1]$$

(d) Find $g(x)g(x) - gg(x)$, giving your answer in the form $ax^2 + bx + c$.

..... [4]

(e) Find $hh(x)$, giving your answer in its simplest form.*Question continues on the next page →*

..... [1]

(f) Find $j(5)$.

..... [1]

(g) Find x when $j^{-1}(x) = 2$. $x =$ [1](h) $j(x) = hg(-12)$ Find the value of x . $x =$ [2]

70 Oct/Nov 2020 · Paper 42 · Q10

(a)(i) $\boxed{10}$ (ii) $g(10) = \boxed{-19}$ (b) $\boxed{\frac{1-x}{2}}$ (c) $\frac{1}{x} = 2 \Rightarrow \boxed{x = \frac{1}{2}}$

(d) $(1-2x)^2 - (1-2(1-2x)) = 1-4x+4x^2-4x+1 = \boxed{4x^2-8x+2}$

(e) $\boxed{x}$ (f) $\boxed{3125}$ (g) $x = j(2) = \boxed{25}$ (h) $hg(-12) = h(25) = \frac{1}{25} = 5^{-2} \Rightarrow \boxed{x = -2}$

71

Oct/Nov 2020 · Paper 43 · Q10

$f(x) = 4 - 3x$

$g(x) = x^2 + x$

$h(x) = 3^x$

(a) Find $fh(2)$.

..... [2]

(b) Find $f^{-1}(x)$. $f^{-1}(x) =$ [2]

(c) Simplify.

(i) $f(1 - 2x)$

..... [2]

(ii) $gf(x) - 9g(x)$

..... [4]

(d) $\frac{1}{h(x)} = 9^{kx}$

Find the value of k . $k =$ [2]

71 Oct/Nov 2020 · Paper 43 · Q10

- (a) $h(2) = 9$, $f(9) = \boxed{-23}$ (b) $\boxed{\frac{4-x}{3}}$ (c)(i) $4 - 3(1 - 2x) = \boxed{6x + 1}$
(ii) $(4 - 3x)^2 + (4 - 3x) - 9x^2 - 9x = \boxed{20 - 36x}$
(d) $\frac{1}{3^x} = 3^{-x} = 9^{kx} = 3^{2kx} \Rightarrow 2k = -1 \Rightarrow \boxed{k = -\frac{1}{2}}$

72

May/June 2020 • Paper 42 • Q6

$f(x) = 3x + 2$

$g(x) = x^2 + 1$

$h(x) = 4^x$

(a) Find $h(4)$.

..... [1]

(b) Find $fg(1)$.

..... [2]

(c) Find $gf(x)$ in the form $ax^2 + bx + c$.

..... [3]

(d) Find x when $f(x) = g(7)$. $x =$ [2](e) Find $f^{-1}(x)$. $f^{-1}(x) =$ [2](f) Find $\frac{g(x)}{f(x)} + x$.Give your answer as a single fraction, in terms of x , in its simplest form.*Question continues on the next page →*

..... [3]

(g) Find x when $h^{-1}(x) = 2$.

$x =$ [1]

72 May/June 2020 · Paper 42 · Q6

(a) $4^4 = \boxed{256}$ (b) $g(1) = 2$, $fg(1) = \boxed{8}$ (c) $(3x + 2)^2 + 1 =$

$\boxed{9x^2 + 12x + 5}$

(d) $g(7) = 50$: $3x + 2 = 50 \Rightarrow \boxed{x = 16}$

(e) $\boxed{f^{-1}(x) = \frac{x-2}{3}}$

(f) $\frac{x^2 + 1}{3x + 2} + \frac{x(3x + 2)}{3x + 2} = \boxed{\frac{4x^2 + 2x + 1}{3x + 2}}$

(g) $x = h(2) = \boxed{16}$

73

May/June 2020 · Paper 43 · Q11

$$f(x) = 7x - 4$$

$$g(x) = \frac{2x}{x-3}, x \neq 3$$

$$h(x) = x^2$$

(a) Find $g(6)$.

..... [1]

(b) Find $fg(4)$.

..... [2]

(c) Find $fh(x)$.

..... [1]

(d) Find $\frac{f(x)}{2} + g(x)$.

Give your answer as a single fraction, in terms of x , in its simplest form.

..... [3]

(e) Find the value of x when $f(x+2) = -11$.

Question continues on the next page →

$$x = \dots\dots\dots [2]$$

- (f) Find the values of p that satisfy $h(p) = p$.

$$\dots\dots\dots [2]$$

73 May/June 2020 · Paper 43 · Q11

(a) $\frac{12}{3} = \boxed{4}$ (b) $g(4) = 8, f(8) = \boxed{52}$ (c) $\boxed{7x^2 - 4}$

(d) $\frac{(7x-4)(x-3)+4x}{2(x-3)} = \boxed{\frac{7x^2-21x+12}{2(x-3)}}$

(e) $7(x+2)-4 = -11 \Rightarrow 7x = -21 \Rightarrow \boxed{x = -3}$ (f) $p^2 = p \Rightarrow$

$\boxed{p = 0 \text{ or } 1}$

74

Oct/Nov 2015 • Paper 43 • Q9

$$f(x) = 2x - 1$$

$$g(x) = \frac{1}{x}, \quad x \neq 0$$

$$h(x) = 2^x$$

(a) Find $h(3)$.

Answer(a) [1]

(b) Find $fg(0.5)$.

Answer(b) [2]

(c) Find $f^{-1}(x)$.

Answer(c) $f^{-1}(x) =$ [2]

(d) Find $ff(x)$, giving your answer in its simplest form.

Answer(d) [2]

(e) Find $(f(x))^2 + 6$, giving your answer in its simplest form.

Question continues on the next page →

Answer(e) [2]

(f) Simplify $hh^{-1}(x)$.

Answer(f) [1]

(g) Which of the following statements is true?

$$f^{-1}(x) = f(x)$$

$$g^{-1}(x) = g(x)$$

$$h^{-1}(x) = h(x)$$

Answer(g) [1]

(h) Use two of the functions $f(x)$, $g(x)$ and $h(x)$ to find the composite function which is equal to $2^{x+1} - 1$.

Answer(h) [1]

74 Oct/Nov 2015 · Paper 43 · Q9

- (a) $\boxed{8}$ (b) $g(0.5) = 2, f(2) = \boxed{3}$ (c) $\boxed{\frac{x+1}{2}}$ (d) $\boxed{4x-3}$ (e) $(2x-1)^2 + 6 = \boxed{4x^2 - 4x + 7}$
- (f) $\boxed{x}$ (g) $\boxed{g^{-1}(x) = g(x)}$ (h) $f(h(x)) = 2 \cdot 2^x - 1 = 2^{x+1} - 1$: $\boxed{fh(x)}$

75

Oct/Nov 2015 • Paper 4 • Q9

$f(x) = 2x + 5$

$g(x) = 2^x$

$h(x) = 7 - 3x$

(a) Find**(i)** $f(3)$,*Answer(a)(i)* [1]**(ii)** $gg(3)$.*Answer(a)(ii)* [2]**(b)** Find $f^{-1}(x)$.*Answer(b)* $f^{-1}(x) =$ [2]**(c)** Find $fh(x)$, giving your answer in its simplest form.*Answer(c)* [2]**(d)** Find the integer values of x which satisfy this inequality.

$$1 < f(x) \leq 9$$

Answer(d) [3]

75 *Oct/Nov 2015 · Paper 4 · Q9*

(a)(i) $\boxed{11}$ (ii) $g(8) = \boxed{256}$ (b) $\boxed{\frac{x-5}{2}}$ (c) $2(7-3x)+5 = \boxed{19-6x}$
(d) $1 < 2x+5 \leq 9 \Rightarrow -2 < x \leq 2$: $\boxed{-1, 0, 1, 2}$

76

May/June 2015 · Paper 43 · Q10

$$f(x) = 2x - 1$$

$$g(x) = x^2 + x$$

$$h(x) = \frac{2}{x}, x \neq 0$$

- (a) Find $ff(3)$.

Answer(a) [2]

- (b) Find $gf(x)$, giving your answer in its simplest form.

Answer(b) [3]

- (c) Find $f^{-1}(x)$.

Answer(c) $f^{-1}(x) =$ [2]

- (d) Find $h(x) + h(x + 2)$, giving your answer as a single fraction.

Answer(d) [4]

76 May/June 2015 · Paper 43 · Q10

(a) $f(5) = \boxed{9}$ (b) $(2x - 1)^2 + (2x - 1) = \boxed{4x^2 - 2x}$ (c) $\boxed{\frac{x+1}{2}}$

(d) $\frac{2}{x} + \frac{2}{x+2} = \boxed{\frac{4x+4}{x(x+2)}}$

77

May/June 2014 · Paper 41 · Q10

(a) $f(x) = 2x - 3$ $g(x) = \frac{1}{x+1} + 2$ $h(x) = 3^x$

(i) Work out $f(4)$.

Answer(a)(i) [1]

(ii) Work out $fh(-1)$.

Answer(a)(ii) [2]

(iii) Find $f^{-1}(x)$, the inverse of $f(x)$.

Answer(a)(iii) $f^{-1}(x) =$ [2]

(iv) Find $ff(x)$ in its simplest form.

Answer(a)(iv) $ff(x) =$ [2]

(v) Show that the equation $f(x) = g(x)$ simplifies to $2x^2 - 3x - 6 = 0$.

Answer(a)(v)

Question continues on the next page →

[3]

- (vi) Solve the equation $2x^2 - 3x - 6 = 0$.

Give your answers correct to 2 decimal places.
Show all your working.

Answer(a)(vi) $x = \dots\dots\dots$ or $x = \dots\dots\dots$ [4]

77 May/June 2014 · Paper 41 · Q10

(a)(i) $\boxed{5}$ (ii) $h(-1) = \frac{1}{3}, f(\frac{1}{3}) = \boxed{-2\frac{1}{3}}$ (iii) $\boxed{\frac{x+3}{2}}$ (iv) $\boxed{4x-9}$

(v) $2x-3 = \frac{1}{x+1} + 2 \Rightarrow (2x-5)(x+1) = 1 \Rightarrow 2x^2 - 3x - 6 = 0 \checkmark$

(vi) $x = \frac{3 \pm \sqrt{57}}{4}$: $\boxed{2.64 \text{ or } -1.14}$ (b) $\frac{(x-1)(x-2)}{(x+5)(x-2)} = \boxed{\frac{x-1}{x+5}}$

78

May/June 2014 • Paper 42 • Q10

$$f(x) = \frac{1}{x}, \quad x \neq 0 \qquad g(x) = 1 - x \qquad h(x) = x^2 + 1$$

- (a) Find $fg\left(\frac{1}{2}\right)$.

Answer(a) [2]

- (b) Find $g^{-1}(x)$, the inverse of $g(x)$.

Answer(b) $g^{-1}(x) =$ [1]

- (c) Find $hg(x)$, giving your answer in its simplest form.

Answer(c) $hg(x) =$ [3]

- (d) Find the value of x when $g(x) = 7$.

Answer(d) $x =$ [1]

- (e) Solve the equation $h(x) = 3x$.

Show your working and give your answers correct to 2 decimal places.

Answer(e) $x =$ or $x =$ [4]

- (f) A function $k(x)$ is its own inverse when $k^{-1}(x) = k(x)$.

For which of the functions $f(x)$, $g(x)$ and $h(x)$ is this true?

Answer(f) [1]

78 *May/June 2014 · Paper 42 · Q10*

(a) $g(\frac{1}{2}) = \frac{1}{2}$, $f(\frac{1}{2}) = \boxed{2}$ (b) $\boxed{1-x}$ (c) $(1-x)^2 + 1 = \boxed{x^2 - 2x + 2}$

(d) $\boxed{-6}$

(e) $x^2 - 3x + 1 = 0$: $\boxed{0.38 \text{ or } 2.62}$ (f) $\boxed{f(x) \text{ and } g(x)}$

79

May/June 2014 • Paper 43 • Q9

$$f(x) = 4 - 3x$$

$$g(x) = 3^{-x}$$

- (a) Find $f(2x)$ in terms of x .

Answer(a) $f(2x) = \dots\dots\dots$ [1]

- (b) Find $ff(x)$ in its simplest form.

Answer(b) $ff(x) = \dots\dots\dots$ [2]

- (c) Work out $gg(-1)$.
Give your answer as a fraction.

Answer(c) $\dots\dots\dots$ [3]

- (d) Find $f^{-1}(x)$, the inverse of $f(x)$.

Answer(d) $f^{-1}(x) = \dots\dots\dots$ [2]

- (e) Solve the equation $gf(x) = 1$.

Answer(e) $x = \dots\dots\dots$ [3]

79 May/June 2014 · Paper 43 · Q9

(a) $4 - 6x$ (b) $4 - 3(4 - 3x) = 9x - 8$ (c) $g(-1) = 3, g(3) = 3^{-3} =$
 $\frac{1}{27}$

(d) $\frac{4 - x}{3}$ (e) $3^{-(4-3x)} = 1 \Rightarrow 3x - 4 = 0 \Rightarrow x = \frac{4}{3}$

80

Oct/Nov 2013 · Paper 41 · Q6(b)

$$g(x) = 1 - 2x \quad h(x) = x^2 - 1$$

(i) Find $gh(3)$.

Answer(b)(i) [2]

(ii) Find $g^{-1}(x)$.

Answer(b)(ii) $g^{-1}(x) =$ [2]

(iii) Solve the equation $h(x) = 3$.

Answer(b)(iii) $x =$ or $x =$ [3]

(iv) Solve the equation $g(3x) = 2x$.

Answer(b)(iv) $x =$ [3]

80 *Oct/Nov 2013 · Paper 41 · Q6(b)*

(i) $h(3) = 8$, $g(8) = \boxed{-15}$ (ii) $\boxed{\frac{1-x}{2}}$ (iii) $x^2 = 4 \Rightarrow \boxed{x = 2 \text{ or } -2}$
(iv) $1 - 6x = 2x \Rightarrow \boxed{x = \frac{1}{8}}$

81

May/June 2013 • Paper 41 • Q9

$$f(x) = x^2 + x - 3$$

$$g(x) = 2x + 7$$

$$h(x) = 2^x$$

- (a) Solve the equation $f(x) = 0$.
Show all your working and give your answers correct to 2 decimal places.

Answer(a) $x = \dots\dots\dots$ or $x = \dots\dots\dots$ [4]

- (b) $fg(x) = px^2 + qx + r$

Find the values of p , q and r .

Question continues on the next page →

Answer(b) $p = \dots\dots\dots$

$q = \dots\dots\dots$

$r = \dots\dots\dots$ [3]

(c) Find $g^{-1}(x)$.

Answer(c) $g^{-1}(x) = \dots\dots\dots$ [2]

(d) Find x when $h(x) = 0.25$.

Answer(d) $x = \dots\dots\dots$ [1]

(e) Find $hhh(3)$.

Give your answer in standard form, correct to 4 significant figures.

Question continues on the next page $\rightarrow$

Answer(e) [4]

81 May/June 2013 · Paper 41 · Q9

(a) $x = \frac{-1 \pm \sqrt{13}}{2}$: $\boxed{1.30 \text{ or } -2.30}$ (b) $(2x + 7)^2 + (2x + 7) - 3 = 4x^2 + 30x + 53$: $\boxed{p = 4, q = 30, r = 53}$

(c) $\boxed{\frac{x-7}{2}}$ (d) $2^x = \frac{1}{4} \Rightarrow \boxed{x = -2}$ (e) $h(3) = 8, h(8) = 256, h(256) = 2^{256} = \boxed{1.158 \times 10^{77}}$

82

2013 • Paper 4 • Q8

$$f(x) = 4x + 3 \qquad g(x) = \frac{7}{x+1} \quad (x \neq -1) \qquad h(x) = x^2 + 5x$$

(a) Work out

(i) $h(-3)$,

Answer(a)(i) [1]

(ii) $hg(13)$.

Answer(a)(ii) [2]

(b) Find $f^{-1}(x)$.Answer(b) $f^{-1}(x) =$ [2](c) (i) Solve the equation $f(x) = 23$.Answer(c)(i) $x =$ [2](ii) Solve the equation $h(x) = 7$.

Show all your working and give your answers correct to 2 decimal places.

Question continues on the next page →

Answer(c)(ii) $x = \dots\dots\dots$ or $x = \dots\dots\dots$ [5]

82 2013 · Paper 4 · Q8

- (a)(i) $\boxed{-6}$ (ii) $g(13) = 0.5, h(0.5) = \boxed{2.75}$ (b) $\boxed{\frac{x-3}{4}}$
- (c)(i) $4x + 3 = 23 \Rightarrow \boxed{x = 5}$ (ii) $x^2 + 5x - 7 = 0$: $\boxed{1.14 \text{ or } -6.14}$

83

May/June 2012 • Paper 41 • Q9

$f(x) = 3x + 5$

$g(x) = 7 - 2x$

$h(x) = x^2 - 8$

(a) Find

(i) $f(3)$,

Answer(a)(i) [1]

(ii) $g(x - 3)$ in terms of x in its simplest form,

Answer(a)(ii) [2]

(iii) $h(5x)$ in terms of x in its simplest form.

Answer(a)(iii) [1]

(b) Find the inverse function $g^{-1}(x)$.Answer(b) $g^{-1}(x) =$ [2](c) Find $hf(x)$ in the form $ax^2 + bx + c$.Answer(c) $hf(x) =$ [3](d) Solve the equation $ff(x) = 83$.Answer(d) $x =$ [3](e) Solve the inequality $2f(x) < g(x)$.

Answer(e) [3]

83 *May/June 2012 · Paper 41 · Q9*

(a)(i) $\boxed{14}$ (ii) $7 - 2(x - 3) = \boxed{13 - 2x}$ (iii) $\boxed{25x^2 - 8}$ (b) $\boxed{\frac{7-x}{2}}$

(c) $(3x + 5)^2 - 8 = \boxed{9x^2 + 30x + 17}$ (d) $3(3x + 5) + 5 = 83 \Rightarrow \boxed{x = 7}$

(e) $6x + 10 < 7 - 2x \Rightarrow \boxed{x < -\frac{3}{8}}$

84

May/June 2012 • Paper 43 • Q9

$$f(x) = 1 - 2x$$

$$g(x) = \frac{1}{x}, \quad x \neq 0$$

$$h(x) = x^3 + 1$$

(a) Find the value of

(i) $gf(2)$,

Answer(a)(i) [2]

(ii) $h(-2)$.

Answer(a)(ii) [1]

(b) Find $fg(x)$.

Write your answer as a single fraction.

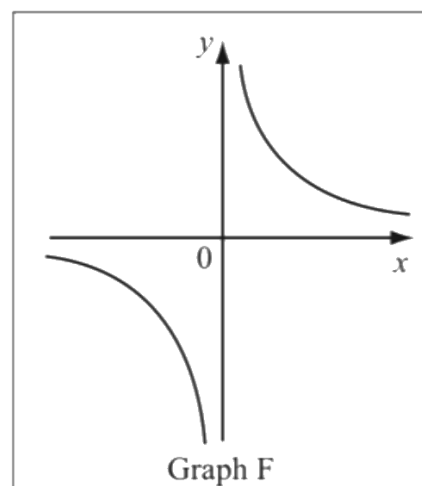
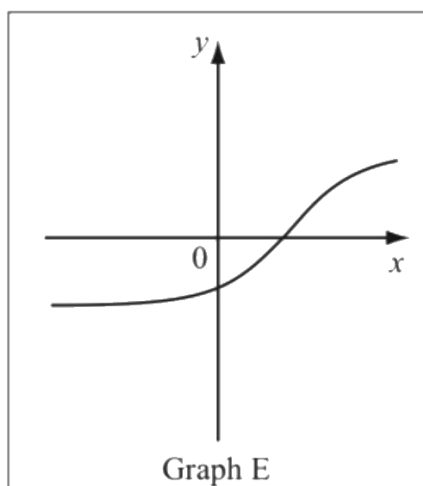
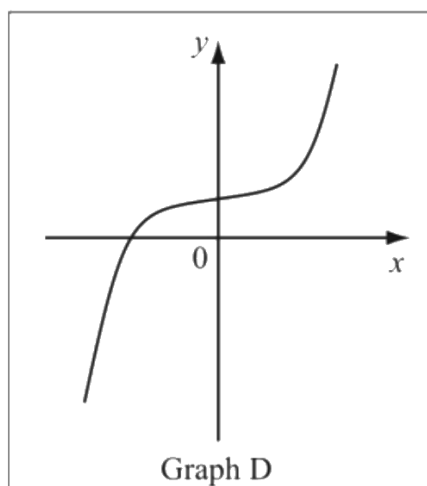
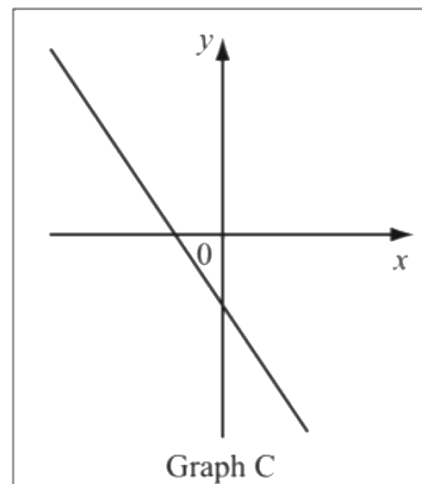
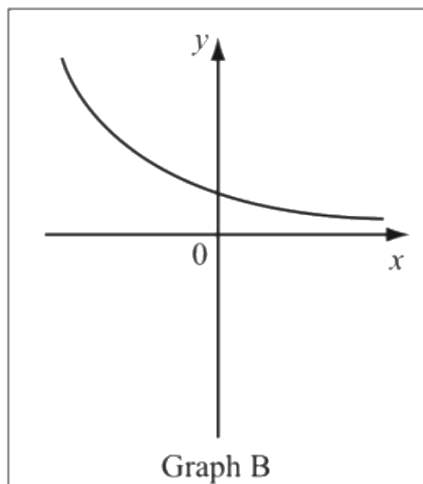
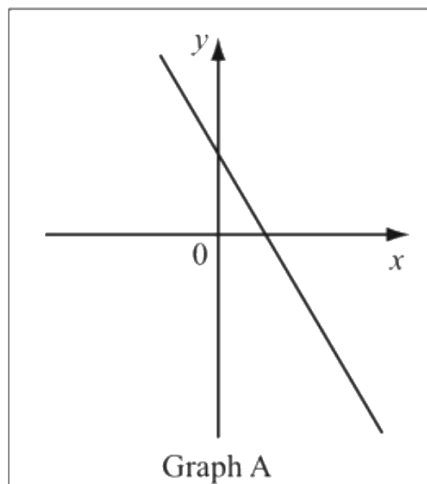
Answer(b) $fg(x) =$ [2]

(c) Find $h^{-1}(x)$, the inverse of $h(x)$.

Question continues on the next page →

Answer(c) $h^{-1}(x) = \dots\dots\dots$ [2]

(d) Write down which of these sketches shows the graph of each of $y = f(x)$, $y = g(x)$ and $y = h(x)$.



Answer(d) $y = f(x)$ Graph $\dots\dots\dots$

$y = g(x)$ Graph $\dots\dots\dots$

$y = h(x)$ Graph $\dots\dots\dots$ [3]

(e) $k(x) = x^5 - 3$

Solve the equation $k^{-1}(x) = 2$.

Question continues on the next page $\rightarrow$

Answer(e) x = [2]

84 *May/June 2012 · Paper 43 · Q9*

(a)(i) $f(2) = -3$, $g(-3) = \boxed{-\frac{1}{3}}$ (ii) $\boxed{-7}$ (b) $1 - \frac{2}{x} = \boxed{\frac{x-2}{x}}$ (c)

$\boxed{\sqrt[3]{x-1}}$

(d) $\boxed{f: A, g: F, h: D}$ (e) $x = k(2) = 32 - 3 = \boxed{29}$

85

Oct/Nov 2011 • Paper 43 • Q8

$f(x) = x^2 + x - 1$

$g(x) = 1 - 2x$

$h(x) = 3^x$

- (a) Find the value of
- $hg(-2)$
- .

Answer(a) [2]

- (b) Find
- $g^{-1}(x)$
- .

Answer(b) $g^{-1}(x) =$ [2]

- (c) Solve the equation
- $f(x) = 0$
- .
-
- Show all your working and give your answers correct to 2 decimal places.

Answer(c) $x =$ or $x =$ [4]

- (d) Find
- $fg(x)$
- .
-
- Give your answer in its simplest form.

Answer(d) $fg(x) =$ [3]

- (e) Solve the equation
- $h^{-1}(x) = 2$
- .

Answer(e) $x =$ [1]

85 *Oct/Nov 2011 · Paper 43 · Q8*

(a) $g(-2) = 5$, $3^5 = \boxed{243}$ (b) $\boxed{\frac{1-x}{2}}$ (c) $x = \frac{-1 \pm \sqrt{5}}{2}$: $\boxed{0.62 \text{ or } -1.62}$

(d) $(1-2x)^2 + (1-2x) - 1 = \boxed{4x^2 - 6x + 1}$ (e) $x = h(2) = \boxed{9}$

86

May/June 2011 • Paper 43 • Q9

$$f(x) = 3x + 1$$

$$g(x) = (x + 2)^2$$

(a) Find the values of

(i) $gf(2)$,

Answer(a)(i) [2]

(ii) $ff(0.5)$.

Answer(a)(ii) [2]

(b) Find $f^{-1}(x)$, the inverse of $f(x)$.

Answer(b) [2]

(c) Find $fg(x)$.

Give your answer in its simplest form.

Answer(c) [2]

Question continues on the next page →

- (d) Solve the equation $x^2 + f(x) = 0$.

Show all your working and give your answers correct to 2 decimal places.

Answer(d) $x =$ or $x =$ [4]

86 *May/June 2011 · Paper 43 · Q9*

(a)(i) $f(2) = 7$, $g(7) = \boxed{81}$ (ii) $f(0.5) = 2.5$, $f(2.5) = \boxed{8.5}$ (b) $\boxed{\frac{x-1}{3}}$

(c) $3(x+2)^2 + 1 = \boxed{3x^2 + 12x + 13}$ (d) $x^2 + 3x + 1 = 0$: $\boxed{-0.38 \text{ or } -2.62}$

87

Oct/Nov 2010 • Paper 41 • Q8(a)

(a) $f(x) = 2x - 1$ $g(x) = x^2$

Work out

(i) $f(2)$,

Answer(a)(i) [1]

(ii) $g(-2)$,

Answer(a)(ii) [1]

(iii) $ff(x)$ in its simplest form,

Answer(a)(iii) $ff(x) =$ [2]

(iv) $f^{-1}(x)$, the inverse of $f(x)$,

Answer(a)(iv) $f^{-1}(x) =$ [2]

(v) x when $gf(x) = 4$.

Answer(a)(v) $x =$ or $x =$ [4]

87 Oct/Nov 2010 · Paper 41 · Q8(a)

(i) $\boxed{3}$ (ii) $\boxed{4}$ (iii) $2(2x - 1) - 1 = \boxed{4x - 3}$ (iv) $\boxed{\frac{x+1}{2}}$

(v) $(2x - 1)^2 = 4 \Rightarrow 2x - 1 = \pm 2$: $\boxed{x = 1\frac{1}{2} \text{ or } -\frac{1}{2}}$

88

Oct/Nov 2010 • Paper 42 • Q2

$$f(x) = 6 + x^2$$

$$g(x) = 4x - 1$$

(a) Find

(i) $g(3)$,

Answer(a)(i) [1]

(ii) $f(-4)$.

Answer(a)(ii) [1]

(b) Find the inverse function $g^{-1}(x)$.Answer(b) $g^{-1}(x) =$ [2](c) Find $fg(x)$ in its simplest form.Answer(c) $fg(x) =$ [3](d) Solve the equation $gg(x) = 3$.Answer(d) $x =$ [3]

88 *Oct/Nov 2010 · Paper 42 · Q2*

(a)(i) $\boxed{11}$ (ii) $\boxed{22}$ (b) $\boxed{\frac{x+1}{4}}$ (c) $6 + (4x - 1)^2 = \boxed{16x^2 - 8x + 7}$

(d) $4(4x - 1) - 1 = 3 \Rightarrow 16x = 8 \Rightarrow \boxed{x = \frac{1}{2}}$